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 $\wedge tx\_status[t] = \text{"Active"}$ 
 $\wedge pc[t] \leq Len(Workload[t].ops)$ 
 $\wedge LET\ op \triangleq Workload[t].ops[pc[t]]$ 
  IN  IF  $op.type = \text{"R"}$ 
    THEN LET  $is\_in\_buffer \triangleq tx\_buffer[t][op.key] \neq \langle \rangle$ 
       $latest\_ver \triangleq MaxVersion(store[op.key], tx\_snapshot[t])$ 
       $read\_val \triangleq$ 
        IF  $is\_in\_buffer$ 
          THEN  $tx\_buffer[t][op.key][1]$ 
          ELSE  $latest\_ver.val$ 
    IN   $\wedge op\_seq' =$ 
       $[op\_seq\ EXCEPT$ 
         $![t] =$ 
         $Append(@, [type \mapsto \text{"R"}, key \mapsto op.key, val \mapsto read\_val])]$ 
       $\wedge client\_dep' =$ 
        IF  $\neg is\_in\_buffer \wedge$ 
           $latest\_ver.cts > client\_dep[Workload[t].session]$ 
        THEN  $[client\_dep\ EXCEPT$ 
           $![Workload[t].session] =$ 
           $latest\_ver.cts]$ 
        ELSE  $client\_dep$ 
       $\wedge tx\_checkset' =$ 
        IF  $Workload[t].level = \text{"SER"}$ 
        THEN  $[tx\_checkset\ EXCEPT$ 
           $![t] =$ 
           $tx\_checkset[t] \cup \{op.key\}]$ 
        ELSE  $tx\_checkset$ 
       $\wedge pc' = [pc\ EXCEPT\ ![t] = pc[t] + 1]$ 
       $\wedge UNCHANGED\ \langle client\_last\_commit,$ 
         $router\_ct,$ 
         $shard\_ct,$ 
         $tx\_status,$ 
         $tx\_snapshot,$ 
         $tx\_buffer,$ 
         $tx\_commit\_time,$ 
         $store$ 
       $\rangle$ 
    ELSE  $op.type = \text{"W"}$ 
       $\wedge tx\_buffer' = [tx\_buffer\ EXCEPT\ ![t][op.key] = \langle op.val \rangle]$ 
       $\wedge tx\_checkset' =$ 
        IF  $Workload[t].level \in \{\text{"PSI"}, \text{"SI"}, \text{"SER"}\}$ 
        THEN  $[tx\_checkset\ EXCEPT$ 
           $![t] =$ 
           $tx\_checkset[t] \cup \{op.key\}]$ 
        ELSE  $tx\_checkset$ 

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